**Project Design Phase-II**
**Solution Requirements (Functional & Non-functional)**

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|---|---|
| Date | 28-10-2026 |
| Team ID | EN25CS3010377 |
| Project Name | EV Market Performance and Insights |
| Maximum Marks | 4 Marks |

**Functional Requirements:**
Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Data Access & Integration | Connect Tableau to the EV market dataset (CSV / Excel) covering vehicle sales, price, battery specifications, driving range, manufacturer, manufacturing country, customer ratings, market region and CO2 metrics; relate or join the tables on common fields. | High |
| 2 | Data Processing | Clean and standardize the data (data types, null values, duplicates, category names); assign geographic roles to country and region; prepare fields such as battery type, drivetrain, vehicle category and model year for analysis. | High |
| 3 | Business Analysis | Calculate the KPIs and measures: total EV sales, average vehicle price, average vehicle range, CO2 saved (tons per year), 3-year retained value, customer rating, battery cycle life, maximum DC charging power and energy efficiency (Wh per km); aggregate them by category, manufacturer, battery type, country, region and year. | High |
| 4 | Business Insights & Analytical Outputs | Provide two interactive dashboards. Dashboard 1: KPI tiles, production output by country and category, top manufacturers by customer review and sales units, driving range distribution, and 3-year retained value. Dashboard 2: sales trend over time, battery technology performance, charging vs efficiency, and regional CO2 saved by drivetrain. | High |
| 5 | User Interaction & Analysis | Let users filter by vehicle category through the shared legend, highlight and sort data, use tooltips, and compare categories, manufacturers, battery technologies, countries and regions across the charts. | Medium |

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| 6 | Data Update & Reporting | Refresh the data source so that all worksheets and dashboards update, and make the dashboards shareable and exportable (image / PDF) for executive and stakeholder reporting. | Medium |

**Non-functional Requirements:**

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description |
|---|---|---|
| NFR-1 | **Usability** | Dashboards use clear titles, a consistent color legend by vehicle category, tooltips and interactive filters so executives and analysts can understand insights without training. |
| NFR-2 | **Security** | Access to the dataset and workbook is limited to authorized users, and dashboards are shared with view-only permission. The data covers market-level information and contains no personal customer data. |
| NFR-3 | **Reliability** | Dashboards show accurate and consistent values after every refresh; calculated fields are validated against the source data and data connection errors are reported clearly. |
| NFR-4 | **Performance** | Dashboards load and respond to filters within a few seconds by using extracts, aggregated measures and a limited number of marks per chart. |
| NFR-5 | **Availability** | Published dashboards are accessible online to stakeholders whenever needed, and a packaged workbook copy is kept as a backup. |
| NFR-6 | **Scalability** | The design supports more rows, new model years, manufacturers and vehicle categories by refreshing the data extract, without redesigning the dashboards. |